

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-LV-6701-03 Sheet 1 of 1	
OPL Title	Packing Adjustment (PTFE V-Ring)		Discipline	Mechanical	
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE		Area / Unit	6700 - LP SEPARATOR SYSTEM	
Related Interlock	SEQ-6701 (SIL 1)		P&ID	Ref	TJC-LLD-PID-6701
Classification:	☐ Basic Knowledge	☐ Improvement	☒ Trouble Case	Aspect: P-Q-C-D-S-M-E	

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **packing adjustment (ptfe v-ring)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Loose packing leaks hexane at the stem; over-tight packing makes the stem stick and the valve hunt. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate the line and depressurise before touching packing	Isolation & LOTO verified
2	Snug the packing nuts evenly in small increments	Parts match BOM / spec
3	Stroke the valve and check for a leak-free, smooth travel	Torque / clearance to spec
4	Do not over-torque - excessive friction degrades control	No leak / smooth operation
5	Record the packing adjustment in the valve history	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Manual handwheel jack on LV-6701 seized	Corrosion in handwheel thread	Overhauled handwheel assembly, greased thread
LV-6701 passing when closed (Class IV exceeded)	Seat/plug erosion from polymer bottoms	Lapped seat and plug, restored Class IV shut-off
Minor weep at body/bonnet joint	Spiral-wound body gasket relaxed	Renewed body gasket, torqued bonnet bolts

6. KEY LEARNING POINTS

- Loose packing leaks hexane at the stem; over-tight packing makes the stem stick and the valve hunt.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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